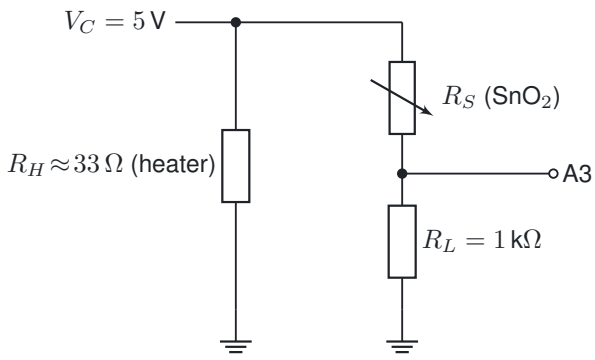
(a) MQ-2 chemiresistor read-out



$$V_{out} = V_C \,\frac{R_L}{R_S + R_L} \;\Rightarrow\; R_S = R_L \,\frac{V_C - V_{out}}{V_{out}}$$

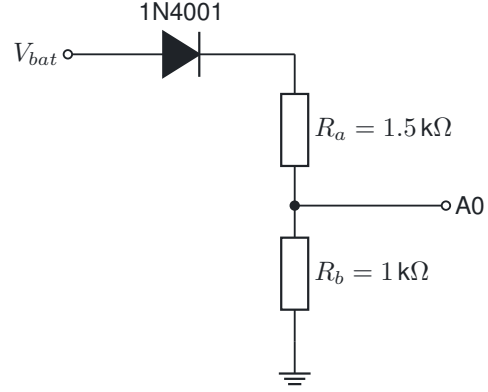
$$V_{out} = \mathrm{ADC} \cdot \frac{5\,\mathrm{V}}{1023}, \quad R_0 = \frac{R_S^{\text{clean air}}}{9.83}$$

$$\log_{10}\frac{R_S}{R_0} = m\,(\log_{10} c - x_0) + y_0$$

$$c = 10^{(\log_{10}(R_S/R_0) - y_0)/m + x_0}\ \text{[ppm]}$$

LPG: $(x_0, y_0, m) = (2.3,\ 0.21,\ -0.47)$

(b) Battery voltage sense



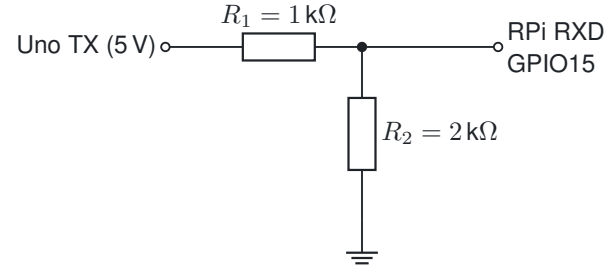
$$V_{A0} = (V_{bat} - V_D)\,\frac{R_b}{R_a + R_b} = 0.4\,(V_{bat} - 0.8\,\mathrm{V})$$

$$V_{bat}\,[\mathrm{cV}] = (\mathrm{ADC} + 65) \cdot 1.2317$$

firmware low-pass: $v_k = 0.92\,v_{k-1} + 0.08\,(\mathrm{ADC} + 65) \cdot 1.2317$

12.6 V full $\rightarrow$ ADC $\approx$ 966 (< 1023, no clipping)

(c) UART level shifting



$$V_{RX} = 5\,\mathrm{V} \cdot \frac{R_2}{R_1 + R_2} = 3.33\,\mathrm{V}$$

$\tau = (R_1 \,\|\, R_2)\,C_{in} \approx 0.67\,\mathrm{k}\Omega \cdot 10\,\mathrm{pF} = 6.7\,\mathrm{ns}$
$\ll$ bit time $1/115\,200 = 8.7\,\mu\mathrm{s}$